

WILLOW HUGHES

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SOFTWARE DEVELOPMENT ENGINEER IN TEST (SDET)

A technically skilled, software development engineer in test with 8+ years of experience and a strong foundation in full-stack software engineering. Leverages deep expertise in Playwright, C#, TypeScript, and enterprise-scale automation to deliver reliable, maintainable, and scalable test frameworks. Excels at architecting UI automation, optimizing end-to-end test suites, and driving CI-aligned quality strategies across complex distributed systems. Brings a proven record of reducing manual testing, elevating engineering standards, and leading teams through technical guidance and cross-functional collaboration. Demonstrates written and oral communication skills, positive attitude, motivation, and commitment to organizational goals in ever-changing environments.

CORE COMPETENCIES

Test Framework Architecture | UI Test Strategy Design | Web Application Architecture | End-to-End Testing | Regression Test Automation
API Testing | Performance Testing | Load Testing | Cross-Browser Testing | Mobile Application Testing | Version Control Workflows
Test Data Management | Continuous Integration | Continuous Delivery | Test Coverage Analysis | Defect Lifecycle Management
Smoke Test Automation | Functional Testing | Authentication & Authorization Testing | Logging & Monitoring Analysis
Agile & Scrum Execution | Code Coverage Tools | Technical Documentation Writing | Team Leadership

TECHNICAL SKILLS

Front-End: Playwright, Selenium, JavaScript, TypeScript, jQuery, HTML, XML, CSS, CSHTML, Bootstrap, React

Back-End: C#, PHP, ASP.NET, .NET Framework, .NET Core, Python, Java, Django, NHibernate, SQL, WebAPI, RESTful Web Services, Database Migrations, Relational Databases, Indexing, Database Profiling

Development Tools: Visual Studio, ReSharper, GitHub, Xcode, IntelliJ IDEA, Android Studio, Chrome Developer Tools, Docker, Microsoft SQL Server Management Studio, Jira, TestRail, BrowserStack

PROFESSIONAL EXPERIENCE

Paycom - Oklahoma City, OK

Software Development Engineer in Test (SDET)

2025 – 2026

Led the architecture and development of Paycom's CRM Playwright automation framework, creating all core components including reusable Table, Navigation, Form abstractions, input handlers, button interactions, verification utilities, and a complete suite of Page Object Models. Coached and onboarded 10+ junior SDETs through hands-on guidance in automation standards and framework architecture. Worked closely with QA analysts, developers, team leads, and managers to align testing with product delivery.

Key Contributions:

- Architected a comprehensive UI automation solution for Paycom's enterprise CRM using C#, Selenium, and Playwright, delivering reusable Page Object Models for Authentication, Navigation, Dropdowns, Forms, and Verification.
- Applied inversion of control, dependency injection, and established test designs to increase scalability and maintainability of CRMs automation codebase.
- Identified client-impacting defect hotspots and created automated test coverage across all relevant CRM submodules, replacing high-volume manual regression steps with maintainable automated tests.
- Delivered a 90% reduction in manual QA involvement and reduced regression cycle time from 3 days to 4 hours through strategic automation improvements.

Nexport Solutions - Oklahoma City, OK

Quality Assurance Tester

2023 – 2024

Executed manual tests for an online learning platform, managing coverage and progress through TestRail and FogBugz to keep engineers, testers, stakeholders, and end users aligned throughout the testing lifecycle.

Key Contributions:

- Performed regression, smoke, and cross-browser testing for a large online learning platform, managing test coverage and execution using TestRail, FogBugz, and BrowserStack.
- Developed structured test plans and test cases to validate point releases, leveraging Postman and Chrome DevTools to verify APIs, UI behavior, and critical workflows.

Nexport Solutions - Oklahoma City, OK

Full Stack Software Engineer

2021 – 2023

Engineered a TypeScript media library tool supporting file uploads with full CRUD functionality, search, and sorting, and integrated into assignment creation, instructor classrooms, and student classrooms to enable media reuse without manual HTML links.

Key Contributions:

- Built and maintained full-stack features using C#, ASP.NET MVC, TypeScript, REST APIs, SQL, and NHibernate, strengthening skills in object-oriented design, ORM patterns, and scalable architecture.
- Improved application reliability by refactoring legacy Monorail components into modern C# MVC, reducing technical debt and enhancing maintainability across multi-module systems.
- Applied advanced SQL querying and database design principles to support efficient data access layers, troubleshoot defects, and optimize performance in relational database environments.
- Increased code quality through NUnit test coverage, multi-platform development experience, and collaborative delivery across distributed engineering teams.

Monte R. Lee and Company - Oklahoma City, OK

Full Stack Software Engineer

2019 – 2021

Developed a real-time data management system for field teams to report wiring units, paired with a Python-based Django web application that supports office reporting and project progress while allowing internal teams to shift toward additional contracts.

Key Contributions:

- Implemented a Kotlin mobile companion application enabling field workers to submit project updates to the office, featuring offline caching and automatic synchronization with a MySQL database across intermittent connectivity scenarios.
- Structured a PHP RESTful API to bridge communication between the mobile application and Django platform, laying the groundwork for compatibility with future third-party service integrations.

University of Oklahoma - Norman, OK

Software Engineer

2018 – 2019

Coordinated development efforts within a 3-person team by partnering directly with the Second Wind manager and applying agile practices to organize work, align requirements, and deliver the solution efficiently.

Key Contributions:

- Created a pay-what-you-want slider tailored for a donation-based coffee shop, simplifying the checkout experience for customers unfamiliar with gratuity-funded pricing while integrating a secure register and payment processing through Square app APIs.

INTERNSHIP EXPERIENCE: Software Engineer Intern, Jardum - Owasso, OK (2016 – 2017)

EDUCATION

Bachelor of Engineering (BE), Computer Science, University of Oklahoma, Norman, OK (2021)
National Career Readiness Certificate, ACT (2014)

KEY PROJECTS

Clay Factoria - Automation Modpack for Hytale | Tech: Java, IntelliJ IDEA (2026)

- Implemented an automation system for Hytale using Clay creatures as a means of item transportation.

Clear Water - Unity Puzzle Game | Tech: C#, Unity (2023)

- Built a Unity puzzle game using maze rotation mechanics and a purity-based scoring system.

Chappie - Personal Home Robot & AI Assistant (2022)

- Developed an autonomous home robot with RGBD navigation, modular hardware control, and AI assistant integrations.

Dialektor - Audio Analysis Tool | Tech: JavaScript, Python, Django, Docker, Kubernetes, PostgreSQL (2021)

- Created an audio recording and ledger system for tracking linguistic and acoustic changes over time.

QuestContinue - ESO Quality-of-Life Mod | Tech: Lua (2021)

- Built a Lua mod that automates quest tracking and supports multi-chapter progression in Elder Scrolls Online.

Cocomo - Software Cost Estimation Model (2020)

- Implemented a project estimation model calculating effort, staffing, and development timelines from code size.

Tri County Tech Mobile App | Tech: JavaScript, jQuery, jQuery Mobile (2016)

- Developed a mobile app for students and staff with calendars, news, menus, and contact features.